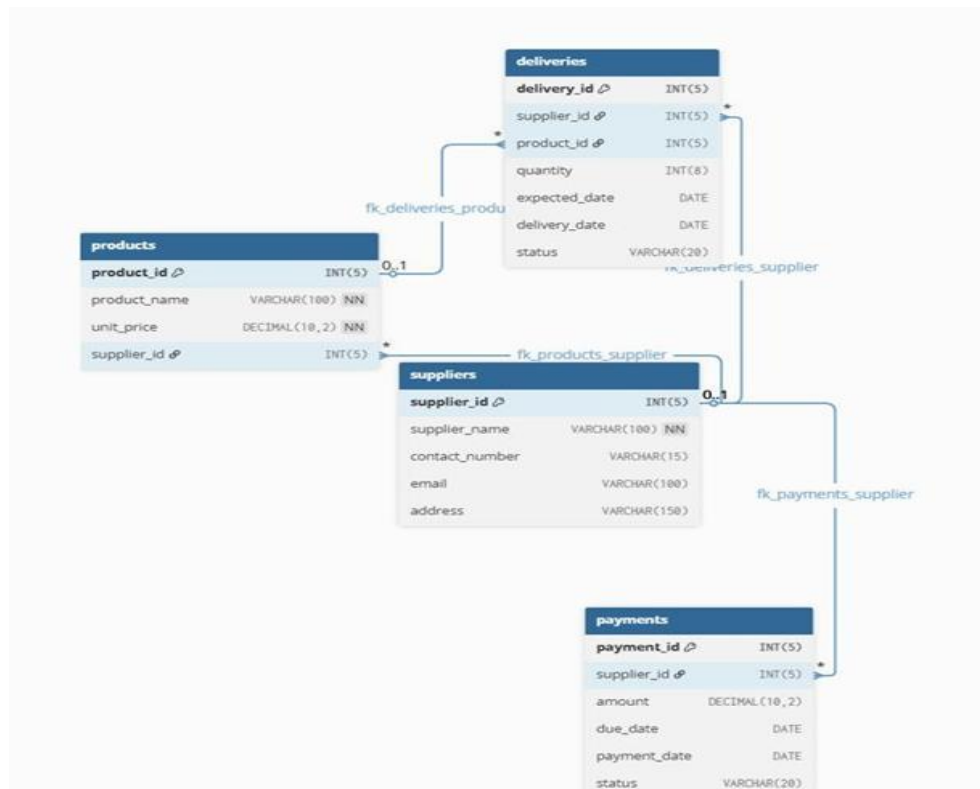


PHASE III: Supplier Payment and Delivery Monitoring System

Project title: Supplier Payment and Delivery Monitoring System

Supplier Payment and Delivery Monitoring System ER diagram



1. Introduction

This document presents a clear logical model for the Supplier Payment and Delivery Monitoring System. It is based on the ER diagram and provides a structured foundation for implementing the physical database.

2. Project Overview

The system is designed to help manage suppliers, products, deliveries, and payments. It ensures accurate tracking of procurement activities and financial transactions.

3. Entity Relationship Analysis

3.1 Core Entities

- Supplier: Stores business partner information.

- Product: Contains details of items purchased from suppliers.
- Delivery: Tracks procurement events, delivery dates, and quantity.
- Payment: Records financial transactions and payment status.

3.2 Relationships

- Supplier to Deliveries: One-to-many relationship.
- Product to Deliveries: One-to-many relationship.
- Supplier to Payments: One-to-many relationship.
- Supplier to Products: One-to-many relationship.

4. Detailed Entity Definitions

4.1 Supplier Entity

Attribute	Data Type	Constraints	Description
supplier_id	INT	PRIMARY KEY	Unique supplier identifier
supplier_name	VARCHAR2(100)	NOT NULL	Supplier name
contact_number	VARCHAR2(20)		Supplier contact number
email	VARCHAR2(100)	UNIQUE	Supplier email address
address	VARCHAR2(255)		Supplier physical address

4.2 Product Entity

Attribute	Data Type	Constraints	Description
product_id	INT	PRIMARY KEY	Unique product identifier
product_name	VARCHAR2(100)	NOT NULL	Name of the product

unit_price	DECIMAL(10,2)	NOT NULL	Price per unit
supplier_id	INT	FOREIGN KEY	Linked supplier

4.3 Delivery Entity

Attribute	Data Type	Constraints	Description
delivery_id	INT	PRIMARY KEY	Unique delivery record
supplier_id	INT	FOREIGN KEY	Supplier linked to delivery
product_id	INT	FOREIGN KEY	Delivered product
quantity	INT	NOT NULL	Amount delivered
expected_date	DATE	NOT NULL	Scheduled delivery date
delivery_date	DATE		Actual delivery date
status	VARCHAR2(30)	NOT NULL	Current delivery status

4.4 Payment Entity

Attribute	Data Type	Constraints	Description
payment_id	INT	PRIMARY KEY	Unique payment identifier
supplier_id	INT	FOREIGN KEY	Supplier receiving payment
amount	DECIMAL(10,2)	NOT NULL	Payment amount
due_date	DATE	NOT NULL	Scheduled payment date
payment_date	DATE		Actual payment date

status	VARCHAR2(30)	NOT NULL	Payment status
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5. Normalization Analysis

The database model adheres to 1NF, 2NF, and 3NF to ensure data integrity and eliminate redundancy.

- 1NF: All attributes are atomic and no repeating groups.
- 2NF: All non-key attributes depend on the full primary key.
- 3NF: No transitive dependencies exist.

6. Additional Constraints and Indexes

CHECK, UNIQUE, and FOREIGN KEY constraints ensure valid data. Indexes improve search and join performance.

7. Data Integrity Rules

- Referential Integrity: All FK constraints enforce valid relationships.
- Entity Integrity: Primary keys uniquely identify records.
- Domain Integrity: CHECK constraints ensure valid value ranges.

8. Conclusion

This logical model provides a robust foundation for implementing the Supplier Payment & Delivery Monitoring System. It ensures complete, secure, and efficient management of business partner transactions.